Ans:- P = 100%., R = 5% , I·T = 20% of 5% = 1% p.a.

Amount = 100% + 5×5% − 5×1% = 120%

= 120% of P = 120% of 8000 = 9600/-.

Ex:-8:- [B.Q:-5]:- Taarun invested 10k on SI @ 8% p.a & another amount on SI @ 20% p.a. At the end of year the total interest on total amount is 12% p.a. The total investment?

Ans:- Assume total amount = x = total principle

1st Amount (princ) = 10k, 2nd Amount (princ) = x − 10k.

[10000 + 8% of 10k] + [x − 10000 + 20% of (x − 10000)] = x + 12% of x

1st A, (or)

8% of 10k + 20% of (x − 10000) = 12% of x

800 + 0.2x − 2000 = 0.12x

0.2x − 1200 = 0.12x

0.2x − 0.12x = 1200 ⇒ 0.08x = 1200

⇒ $x = \frac{1200 \times 100}{8} = 15000$

Ex:-9:- [B.Q:-6]:- Vikram borrows a sum of 1500/- @ beginning of year. After 4 months 2100/- more is borrowed double the previous interest. At the end of year he paid total 416/- as interest. What is 1st rate of interest?

Ans:- 1500/- borrowed for 1 year @ x% p.a (assume) & 2100 borrowed for $\frac{2}{3}$ yrs (1y − 4 months / 8 months) @ 2x% p.a.

x% of 1500 + $\frac{2}{3}$ × 2x % of 2100 = 416

$\frac{x}{100} \times 1500 + \frac{4x}{3 \times 100} \times 2100 = 416$

⇒ 15x + 28x = 43x = 416 ⇒ $x = \frac{416}{43} = \boxed{9.67}$ % p.a

Ex:10:- [B.Q:-9]:- A sum was put @ R% for 4 yrs. Had it been put @ a rate 4% higher than previous, it would fetches 80/- more. What is sum?

Ans:- $4(R+4)\% - 4R\% = 80$

$4R\% + 16\% - 4R\% = 80 \Rightarrow 16\% = 80 \Rightarrow 1\% = 5$

$P = 100\% = 500/-$

(or)

Diff in interests for 4 yrs = $4 \times 4\% = 80 \Rightarrow 1\% = 5/-$

$P = 100\% = 500/-$

Ex:11:- [B.Q:-11]:- A sum of money tripled itself in 16 years. In how many years it becomes 8 times?

Ans:- $P \xrightarrow[16\text{yrs}]{+2P} 3P \xrightarrow[16\text{yrs}]{+2P} 5P \xrightarrow[16\text{yrs}]{+2P} 7P \xrightarrow[8\text{yrs}]{+P} 8P$

$+2P \rightarrow 16\text{yrs}$

$+P \rightarrow 8\text{yrs}$

$P \xrightarrow[+16+16+16+8]{+7P} 8P \Rightarrow$ in (56 yrs)

(or)

$P \xrightarrow[16\text{yr}]{+2P} 3P \Rightarrow P \xrightarrow[?]{+7P} 8P$

$+2P \longrightarrow 16\text{yrs}$

$+7P \longrightarrow ?$

$\Rightarrow \frac{7 \times 16^{8}}{2} = 56\text{ yrs}.$

Ex:12:- [B.Q:-13]:- SI on a certain sum for 2 yrs @ 10% p.a is 90/-. The corresponding CI is?

Ans:- SI for 2 yrs = 90 $\Rightarrow$ 1 year SI = 45/-

CI for 2yrs = SI for 1st years + SI for 2nd year + 10% of SI for 2nd year

$= 45 + 45 + 4.5$

$= 94.5/-$

Ex:-13:- [B.Q:-14] :- Manoj borrowed a certain sum from anuj at a certain rate of Interest for 2 yrs. He lent this sum to Rakesh @ the same rate of Interest compounded annualy for same period. At the end of 2 yrs, he received 4200/- as CI but paid 4000 as SI. Find rate of Interest?

Ans:- 2 yrs S.I paid by manoj to Anuj = 4000/-

$\Rightarrow$ SI for 1year = $\frac{4000}{2}$ = 2000/-

2 yrs's C.I Paid by rakesh to manoj = 4200/-

2 yrs's CI = 1st yr CI = 1st yr SI + 2nd yr CI = 4200

= 2000 + 2nd yr CI = 4200

= 2000 + (1st yr SI + Interest on it) = 4200

$\Rightarrow$ 2nd year CI = (4200 - 2000) = 2200

= 2000 + 200

200 is what % of 2000 = R = 10%.

Ex:-14:- (CI-SI) for 3 yrs @ 20% p.a = 48 $\Rightarrow$ P = ?

[B.Q:-15] $CI - SI = A_2 - A_1 = 48$

$A_2 = P\left(1+\frac{R}{100}\right)^n = (1.2)^3 P$

$A_1 = P + (20\% + 20\% + 20\%)$ of $P = 1.6P$

$A_2 - A_1 = (1.2)^3 P - 1.6P = (1.728 - 1.6)P = 0.128P = 48$

$\Rightarrow P = \frac{48}{0.128} = \frac{48000}{128} = 375/-$

Ex:-15:- (B.Q:-16)

A sum of money is borrowed & paid back in 2 annual installments of 882/- each. allowing 5% p.a Compound Interest. Then the sum borrowed was?